

SYSTEM ADMINISTRATION AND MAINTAINANCE LAB REPORT

Week 6 - Network Configuration and Management

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IT 123 – WEEK 6 LABORATORY REPORT

Network Configuration and Management

Lab Report:	Lab Report 2 (Third Graded Assessment)
Focus:	Parts 1– 4 (Network Fundamentals) and Part 6 (Student Exercise)
Environment:	Ubuntu Server 26.04.1 LTS (VirtualBox)
Servers Used:	Server A (Static IP) and Server B (DHCP)
Submission Folder:	Week6/ (GitHub Repository)

I. Lab Objectives

This lab report documents the configuration, verification, and troubleshooting of network settings on two Ubuntu Server virtual machines. The exercise applies the concepts covered in Parts 1– 4 of the Week 6 laboratory guide, focusing on:

- Configuring a static IP address on one Ubuntu server using Netplan.
- Keeping a second Ubuntu server on DHCP for flexible addressing.
- Verifying connectivity between the two servers and to the internet.
- Troubleshooting and verifying using ping, traceroute, and nslookup.

II. Scenario

The company requires one Ubuntu server to have a static IP address (192.168.1.150) for internal services, while a second server remains on DHCP for flexibility. Both servers must be able to reach each other and access the internet.

Server A	Static IP – 192.168.1.150/24, Gateway 192.168.1.1, DNS 8.8.8.8 and 8.8.4.4
Server B	DHCP – assigned IP (192.168.1.12/24 obtained automatically)

Part 1 – Checking Current Network Configuration

Before any changes were made, the current network settings of the Ubuntu server were checked. This provided a baseline showing the active interface, the default route, and the DNS settings prior to reconfiguration.



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```
catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86213sec preferred_lft 86213sec
    inet6 fd17:625c:f037:2:a00:27ff:fea0:ca67/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86214sec preferred_lft 14214sec
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
192.168.1.1 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
catlyn@ubuntu-server:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search .
catlyn@ubuntu-server:~$
```

Figure 1. Initial network configuration check (ip a, ip route, cat /etc/resolv.conf) performed before any changes were made.

At this point, the enp0s3 interface had an automatically assigned address (10.0.2.15/24) from the default NAT network, with a default route via 10.0.2.2 and a local DNS resolver (127.0.0.53).

Part 2 – Configuring a Static IP Address

Since Ubuntu 20.04+ uses Netplan for network configuration, the next step was to assign a static IP address to the server.



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Reviewing the Interface and Opening the Netplan File

The interface and routing table were checked again using `ip a` and `ip route`, confirming the interface was now on the 192.168.1.0/24 network through DHCP. The Netplan configuration file was then located and opened using `nano` with administrator privileges.

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon Sep 14 02:53:46 AM UTC 2026

System load:            0.09
Usage of /:              28.2% of 24.44GB
Memory usage:           8%
Swap usage:             0%
Processes:              107
Users logged in:        0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea0:ca67

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 192.168.1.7/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 86388sec preferred_lft 86388sec
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$ ip route
default via 192.168.1.1 dev enp0s3 proto dhcp src 192.168.1.7 metric 100
192.168.1.0/24 dev enp0s3 proto kernel scope link src 192.168.1.7 metric 100
192.168.1.1 dev enp0s3 proto dhcp scope link src 192.168.1.7 metric 100
catlyn@ubuntu-server:~$ ls /etc/netplan
00-installer-config.yaml
catlyn@ubuntu-server:~$ sudo nano /etc/netplan/00-installer-config.yaml
[sudo: authenticate] Password: *****_
```

Figure 2. Interface and routing details re-checked, and the Netplan configuration file located and opened using `sudo nano`.



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Netplan File Before Editing

When the Netplan configuration file was opened, it was blank and ready for the static IP settings to be entered.

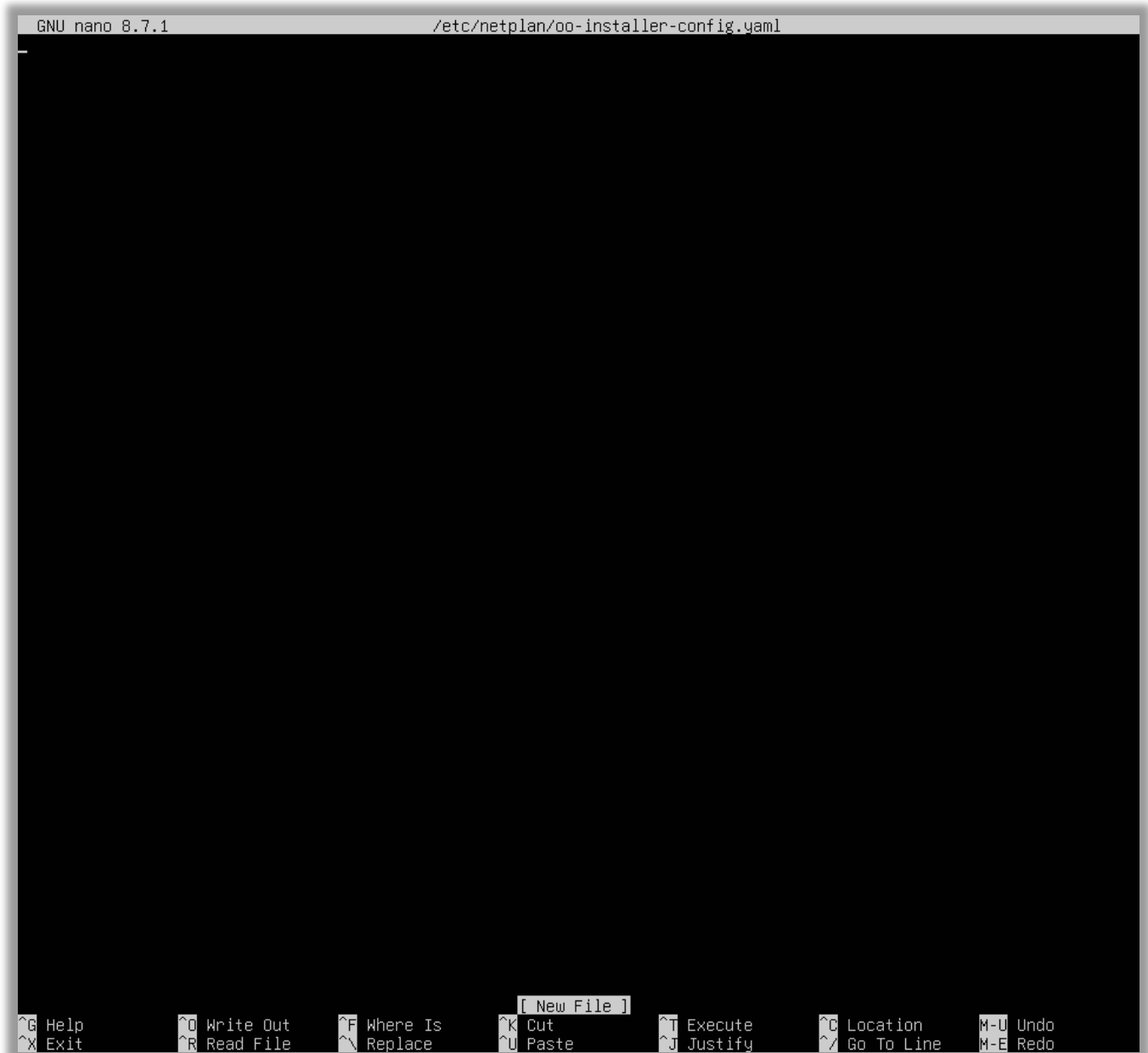


Figure 3. Netplan configuration file shown blank immediately after being opened in nano.



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Entering the Static IP Configuration

The static IP settings were entered into the file. DHCP was turned off (dhcp4: no) and the address 192.168.1.100/24 was assigned, with a gateway of 192.168.1.1 and DNS servers 8.8.8.8 and 8.8.4.4.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml *
network:
version: 2
ethernets:
  enp0s3:
    dhcp4: no
    addresses:
      - 192.168.1.100/24_
    gateway4: 192.168.1.1
    nameservers:
      addresses: [8.8.8.8, 8.8.4.4]
```

GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml *

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

Figure 4. Netplan configuration file after entering the static IP settings for 192.168.1.100/24.



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Applying and Verifying the Static Configuration

The configuration was applied using `sudo netplan apply`. A warning about file permissions and the deprecated `gateway4` key appeared, but it did not prevent the configuration from working. The `ip a` command was used to confirm the interface now had the address `192.168.1.100/24`, and ping tests to `8.8.8.8` and `google.com` both succeeded.

```
catlyn@ubuntu-server:~$ sudo netplan apply

** (configure:1453): WARNING **: 05:04:29.563: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (configure:1453): WARNING **: 05:04:29.563: `gateway4` has been deprecated, use default routes instead. See the 'Default routes' section of the documentation for more details.

** (process:1451): WARNING **: 05:04:30.079: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1451): WARNING **: 05:04:30.081: `gateway4` has been deprecated, use default routes instead. See the 'Default routes' section of the documentation for more details.

** (process:1451): WARNING **: 05:04:30.290: Permissions for /etc/netplan/00-installer-config.yaml are too open. Netplan configuration should NOT be accessible by others.

** (process:1451): WARNING **: 05:04:30.290: `gateway4` has been deprecated, use default routes instead. See the 'Default routes' section of the documentation for more details.
catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 192.168.1.100/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=13.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=14.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=17.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=13.7 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.598/14.882/17.695/1.664 ms
catlyn@ubuntu-server:~$ ping -c 4 google.com
PING google.com (172.217.26.238) 56(84) bytes of data:
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=1 ttl=115 time=14.0 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=2 ttl=115 time=12.5 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=3 ttl=115 time=13.2 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=4 ttl=115 time=24.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 12.515/16.165/24.874/5.056 ms
catlyn@ubuntu-server:~$
```

Figure 5. Result of `sudo netplan apply`, showing the static IP address applied and successful ping tests to `8.8.8.8` and `google.com`.



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Re-verifying After Fixing the Permission Warning

The file permission warning was corrected and the configuration was applied again. This time, only the expected gateway4 deprecation warning appeared. The static IP was confirmed to still be active, and ping tests to 8.8.8.8 and google.com remained successful, before proceeding to Part 3.

```
total 12
drwxr-xr-x  2 root root 4096 Sep 14 05:33 .
drwxr-xr-x 110 root root 4096 Sep 14 02:57 ..
-rw-----  1 root root  192 Sep 14 05:33 00-installer-config.yaml
catlyn@ubuntu-server:~$ sudo netplan apply

** (configure:1461): WARNING **: 05:33:39.879: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1459): WARNING **: 05:33:40.404: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1459): WARNING **: 05:33:40.545: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 192.168.1.100/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=12.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=15.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=14.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=18.3 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 12.933/15.209/18.263/1.963 ms
catlyn@ubuntu-server:~$ ping -c 4 google.com
PING google.com (142.250.205.238) 56(84) bytes of data.
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=1 ttl=116 time=13.7 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=2 ttl=116 time=13.9 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=3 ttl=116 time=14.4 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=4 ttl=116 time=52.1 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.716/23.515/52.081/16.494 ms
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$ sudo nano /etc/netplan/00-installer-config.yaml_
```

Figure 6. Static IP configuration re-verified after correcting the file permission warning.



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Part 3 – Switching Back to DHCP

Editing the Netplan File Back to DHCP

The Netplan configuration file was edited again. The static IP settings were removed and dhcp4: yes was set so the server would resume obtaining its IP address automatically.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml *
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: yes
```

GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml *

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

Figure 7. Netplan configuration file after being edited back to DHCP (dhcp4: yes).



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Applying and Verifying DHCP

The updated configuration was applied using `sudo netplan apply`. Using `ip a`, it was confirmed that the `enp0s3` interface received a new, automatically assigned address (192.168.1.7/24) from the DHCP server.

```
catlyn@ubuntu-server:~$ sudo netplan apply
catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 192.168.1.7/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 86399sec preferred_lft 86399sec
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$
```

Figure 8. Interface showing the new DHCP-assigned IP address (192.168.1.7/24) after switching back to DHCP.

Part 4 – Troubleshooting with CLI Tools

Ping – Testing Connectivity to the Gateway

The connection to the default gateway (192.168.1.1) was tested using `ping` and was successful with 0% packet loss. The `traceroute` tool was then installed to prepare for path tracing.

```
catlyn@ubuntu-server:~$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=2.16 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=19.2 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=4.18 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=8.15 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 2.156/8.409/19.153/6.567 ms
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
```

Figure 9. Successful ping test to the default gateway (192.168.1.1), followed by installation of the `traceroute` tool.



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Installing Traceroute

The traceroute package finished installing without any issues, and no further action, such as a restart, was needed.

```
link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
altname enx080027a0ca67
inet 192.168.1.7/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
    valid_lft 86399sec preferred_lft 86399sec
inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$
catlyn@ubuntu-server:~$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=2.16 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=19.2 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=4.18 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=8.15 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 2.156/8.409/19.153/6.567 ms
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 33
  Download size: 54.4 kB
  Space needed: 167 kB / 17.3 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 traceroute amd64 1:2.1.6-1build1 [54.4 kB]
Fetched 54.4 kB in 1s (74.2 kB/s)
Selecting previously unselected package traceroute.
(Reading database... 99474 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.6-1build1_amd64.deb...
Unpacking traceroute (1:2.1.6-1build1)...
Setting up traceroute (1:2.1.6-1build1)...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.13.1-1build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
catlyn@ubuntu-server:~$
```

Figure 10. Output of the traceroute installation, completed without errors.



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Traceroute – Checking the Packet Path

The traceroute tool was used to trace the path from the server to google.com. This revealed the network hops the packets passed through before reaching the destination.

```
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=2.16 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=19.2 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=4.18 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=8.15 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 2.156/8.409/19.153/6.567 ms
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 33
  Download size: 54.4 kB
  Space needed: 167 kB / 17.3 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 traceroute amd64 1:2.1.6-1build1 [54.4 kB]
Fetched 54.4 kB in 1s (74.2 kB/s)
Selecting previously unselected package traceroute.
(Reading database... 99474 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.6-1build1_amd64.deb...
Unpacking traceroute (1:2.1.6-1build1)...
Setting up traceroute (1:2.1.6-1build1)...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.13.1-1build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
catlyn@ubuntu-server:~$ traceroute google.com
traceroute to google.com (173.194.174.100), 30 hops max, 60 byte packets
 1 * * *
 2 10.56.0.1 (10.56.0.1) 7.275 ms 6.951 ms 7.469 ms
 3 * * *
 4 161.49.4.242.convergeict.com (161.49.4.242) 10.647 ms 12.915 ms 13.026 ms
 5 161.49.11.89.convergeict.com (161.49.11.89) 34.542 ms 161.49.11.81.convergeict.com (161.49.11.81) 36.067 ms 161
.49.5.41.convergeict.com (161.49.5.41) 35.229 ms
 6 142.250.162.104 (142.250.162.104) 29.404 ms 28.655 ms 31.386 ms
 7 td-in-f100.1e100.net (173.194.174.100) 32.343 ms 32.403 ms 60.493 ms
catlyn@ubuntu-server:~$
```

Figure 11. Result of running traceroute google.com, showing the network hops to the destination.



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DNS Resolution and Active Connections

The nslookup command was used to test DNS resolution for google.com, returning both an IPv4 and an IPv6 address (after the dnsutils package was installed). Lastly, ss -tln was used to list the active listening TCP and UDP ports on the server.

```
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
catlyn@ubuntu-server:~$ traceroute google.com
traceroute to google.com (173.194.174.100), 30 hops max, 60 byte packets
 1 * * *
 2 10.56.0.1 (10.56.0.1) 7.275 ms 6.951 ms 7.469 ms
 3 * * *
 4 161.49.4.242.convergeict.com (161.49.4.242) 10.647 ms 12.915 ms 13.026 ms
 5 161.49.11.89.convergeict.com (161.49.11.89) 34.542 ms 161.49.11.81.convergeict.com (161.49.11.81) 36.067 ms 161
.49.5.41.convergeict.com (161.49.5.41) 35.229 ms
 6 142.250.162.104 (142.250.162.104) 29.404 ms 28.655 ms 31.386 ms
 7 td-in-f100.1e100.net (173.194.174.100) 32.343 ms 32.403 ms 60.493 ms
catlyn@ubuntu-server:~$ ns lookup google.com
Command 'ns' not found, but can be installed with:
sudo apt install ns2
catlyn@ubuntu-server:~$ sudo apt install dnsutils -y
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
bind9-dnsutils is already the newest version (1:9.20.24-1ubuntu0.3).
bind9-dnsutils set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.26.78
Name:   google.com
Address: 2404:6800:4017:80a::200e

catlyn@ubuntu-server:~$ ss -tln
Netid  State      Recv-Q    Send-Q      Local Address:Port      Peer Address:Port
udp    UNCONN    0          0          127.0.0.54:53          0.0.0.0:*
udp    UNCONN    0          0          127.0.0.53%lo:53      0.0.0.0:*
udp    UNCONN    0          0          192.168.1.7%enp0s3:68 0.0.0.0:*
udp    UNCONN    0          0          127.0.0.1:323         0.0.0.0:*
udp    UNCONN    0          0          [::1]:323            [::]:*
udp    UNCONN    0          0          [fe80::a00:27ff:fea0:ca67]%enp0s3:546 [::]:*
tcp    LISTEN    0          4096      127.0.0.53%lo:53      0.0.0.0:*
tcp    LISTEN    0          4096      127.0.0.54:53          0.0.0.0:*
tcp    LISTEN    0          4096      0.0.0.0:22            0.0.0.0:*
tcp    LISTEN    0          4096      [::]:22              [::]:*
tcp    LISTEN    0          511      *:80                  *:*
```

Figure 12. nslookup google.com results and ss -tln output showing active listening ports.



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Part 6 – Student Exercise: Server A (Static IP Configuration)

Accessing the Netplan Configuration File

Server A was accessed and the Netplan configuration file was opened using the nano editor with administrative privileges to prepare it for editing.

```
Ubuntu 26.04.1 LTS ubuntu-server tty1
ubuntu-server login: catlyn
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon Sep 14 02:53:46 AM UTC 2026

System load:          0.09
Usage of /:            28.2% of 24.44GB
Memory usage:         8%
Swap usage:           0%
Processes:            107
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea0:ca67

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

catlyn@ubuntu-server:~$ sudo nano /etc/netplan/00-installer-config.yaml
[sudo: authenticate] Password: ****
```

Figure 13. Logging into Server A and opening the Netplan configuration file with sudo nano.



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Before Configuration (Default / DHCP)

Before any changes were made, the configuration file showed the default DHCP setting (dhcp4: yes) for the enp0s3 interface. This confirms the server's original state prior to applying a static IP.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: yes
```

Figure 14. Netplan configuration file before editing, showing the default DHCP setting.

After Configuration (Static IP Applied)

The configuration file was then edited to disable DHCP and assign a static IP address of 192.168.1.150/24, with a gateway of 192.168.1.1 and DNS servers 8.8.8.8 and 8.8.4.4.

```
GNU nano 8.7.1 /etc/netplan/00-installer-config.yaml * S
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.1.150/24
      gateway4: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Figure 15. Netplan configuration file after editing, showing the static IP, gateway, and DNS settings.



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Applying and Verifying the Static IP

The updated configuration was applied using `sudo netplan apply`. A deprecation warning regarding `gateway4` appeared, which is expected behavior in newer Netplan versions and does not prevent the configuration from taking effect. The `ip a` command was then used to confirm that the `enp0s3` interface was successfully assigned the static IP address `192.168.1.150/24`.

```
catlyn@ubuntu-server:~$ ls -la /etc/netplan/
total 12
drwxr-xr-x  2 root root 4096 Sep 14 08:41 .
drwxr-xr-x 110 root root 4096 Sep 14 02:57 ..
-rw-----  1 root root 192 Sep 14 08:41 00-installer-config.yaml
catlyn@ubuntu-server:~$ sudo netplan apply

** (configure:1429): WARNING **: 08:41:46.708: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1427): WARNING **: 08:41:47.228: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:1427): WARNING **: 08:41:47.371: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a0:ca:67 brd ff:ff:ff:ff:ff:ff
    altname enx080027a0ca67
    inet 192.168.1.150/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea0:ca67/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$
```

Figure 16. Applying the Netplan configuration and verifying the static IP address (192.168.1.150) using `ip a`.

Connectivity Test – Ping

From Server A, a ping test was performed toward Server B (192.168.1.12) and toward `google.com` to verify both local network connectivity and internet access. Both tests completed with 0% packet loss.



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```
catlyn@ubuntu-server:~$ ping -c 4 192.168.1.12
PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
64 bytes from 192.168.1.12: icmp_seq=1 ttl=64 time=3.25 ms
64 bytes from 192.168.1.12: icmp_seq=2 ttl=64 time=1.15 ms
64 bytes from 192.168.1.12: icmp_seq=3 ttl=64 time=6.29 ms
64 bytes from 192.168.1.12: icmp_seq=4 ttl=64 time=7.96 ms

--- 192.168.1.12 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.146/4.663/7.963/2.640 ms
catlyn@ubuntu-server:~$ ping -c 4 google.com
PING google.com (142.250.205.238) 56(84) bytes of data.
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=1 ttl=116 time=13.3 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=2 ttl=116 time=13.4 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=3 ttl=116 time=13.5 ms
64 bytes from lcnld-am-in-f14.1e100.net (142.250.205.238): icmp_seq=4 ttl=116 time=12.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 12.665/13.234/13.539/0.339 ms
catlyn@ubuntu-server:~$ _
```

Figure 17. Ping results from Server A to Server B (192.168.1.12) and to google.com, both successful with 0% packet loss.

Connectivity Test – Traceroute

The traceroute tool was installed and used to trace the network path from Server A to google.com. The output shows the sequence of network hops the packets passed through before reaching the destination.

```
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
[sudo: authenticate] Password:
traceroute is already the newest version (1:2.1.6-1build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ traceroute google.com
traceroute to google.com (172.217.26.238), 30 hops max, 60 byte packets
 1 * * *
 2 10.56.0.1 (10.56.0.1) 7.368 ms 6.885 ms 8.837 ms
 3 * * *
 4 161.49.4.242.convergeict.com (161.49.4.242) 12.445 ms 161.49.4.240.convergeict.com (161.49.4.240) 12.099 ms 8.995 ms
 5 161.49.13.151.convergeict.com (161.49.13.151) 20.024 ms * *
 6 161.49.13.183.convergeict.com (161.49.13.183) 14.738 ms * 11.483 ms
 7 * 72.14.214.254 (72.14.214.254) 13.940 ms 14.797 ms
 8 72.14.214.254 (72.14.214.254) 27.323 ms * nrt12s51-in-f14.1e100.net (172.217.26.238) 14.090 ms
catlyn@ubuntu-server:~$ _
```

Figure 18. Traceroute results from Server A to google.com, showing the path taken across intermediate hops.



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DNS Resolution Test – nslookup

The nslookup command was used to test DNS resolution on Server A. The query for google.com successfully returned both an IPv4 and an IPv6 address, confirming that DNS resolution was working correctly.

```
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
[sudo: authenticate] Password:
traceroute is already the newest version (1:2.1.6-1build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ traceroute google.com
traceroute to google.com (172.217.26.238), 30 hops max, 60 byte packets
 1  * * *
 2  10.56.0.1 (10.56.0.1)  7.368 ms  6.885 ms  8.837 ms
 3  * * *
 4  161.49.4.242.convergeict.com (161.49.4.242)  12.445 ms  161.49.4.240.convergeict.com (161.49.4.240)  12.099 ms  8.995 ms
 5  161.49.13.151.convergeict.com (161.49.13.151)  20.024 ms * *
 6  161.49.13.183.convergeict.com (161.49.13.183)  14.738 ms *  11.483 ms
 7  * 72.14.214.254 (72.14.214.254)  13.940 ms  14.797 ms
 8  72.14.214.254 (72.14.214.254)  27.323 ms * nrt12s51-in-f14.1e100.net (172.217.26.238)  14.090 ms
catlyn@ubuntu-server:~$ sudo apt install dnsutils -y
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
bind9-dnsutils is already the newest version (1:9.20.24-1ubuntu0.3).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.27.142
Name:   google.com
Address: 2404:6800:4017:806::200e
catlyn@ubuntu-server:~$ _
```

Figure 19. nslookup results on Server A confirming successful DNS resolution for google.com.

Part 6 – Student Exercise: Server B (DHCP Configuration)

Verifying the DHCP-Assigned IP Address

Server B was left on its default DHCP configuration. After logging in, the ip a command confirmed that the enp0s3 interface automatically received an IP address (192.168.1.12/24) from the DHCP server.



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```
Ubuntu 26.04.1 LTS ubuntu-server tty1

ubuntu-server login: catlyn
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon Sep 14 02:53:46 AM UTC 2026

System load:          0.09
Usage of /:            28.2% of 24.44GB
Memory usage:         8%
Swap usage:           0%
Processes:            107
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea0:ca67

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

catlyn@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
   link/ether 08:00:27:ac:77:85 brd ff:ff:ff:ff:ff:ff
   altname enx080027ac7785
   inet 192.168.1.12/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
       valid_lft 86313sec preferred_lft 86313sec
   inet6 fe80::a00:27ff:feac:7785/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
catlyn@ubuntu-server:~$ _
```

Figure 20. Logging into Server B and verifying the DHCP-assigned IP address using `ip a`.



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Confirming the Netplan Configuration

The Netplan configuration file on Server B was displayed using cat to confirm that dhcp4: yes remained set, verifying that the server was correctly configured to obtain its IP address automatically.

```
catlyn@ubuntu-server:~$ sudo cat /etc/netplan/00-installer-config.yaml
[sudo: authenticate] Password:
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: yes
catlyn@ubuntu-server:~$
```

Figure 21. Netplan configuration file on Server B confirming DHCP is enabled (dhcp4: yes).

Connectivity Test – Ping

From Server B, a ping test was performed toward Server A (192.168.1.150) and toward google.com. Both tests completed successfully with 0% packet loss, confirming connectivity between the two servers and to the internet.

```
catlyn@ubuntu-server:~$ ping -c 4 192.168.1.150
PING 192.168.1.150 (192.168.1.150) 56(84) bytes of data.
64 bytes from 192.168.1.150: icmp_seq=1 ttl=64 time=0.883 ms
64 bytes from 192.168.1.150: icmp_seq=2 ttl=64 time=2.06 ms
64 bytes from 192.168.1.150: icmp_seq=3 ttl=64 time=2.33 ms
64 bytes from 192.168.1.150: icmp_seq=4 ttl=64 time=1.59 ms

--- 192.168.1.150 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.883/1.713/2.325/0.546 ms
catlyn@ubuntu-server:~$ ping -c 4 google.com
PING google.com (142.250.207.46) 56(84) bytes of data.
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=1 ttl=116 time=16.2 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=2 ttl=116 time=17.0 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=3 ttl=116 time=16.3 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=4 ttl=116 time=16.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 16.176/16.452/16.980/0.311 ms
catlyn@ubuntu-server:~$
```

Figure 22. Ping results from Server B to Server A (192.168.1.150) and to google.com, both successful with 0% packet loss.



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Connectivity Test – Traceroute

A traceroute was performed from Server B to google.com to verify the network path and identify the intermediate hops between the server and the destination.

```
catlyn@ubuntu-server:~$ sudo apt install traceroute -y
[sudo: authenticate] Password:
traceroute is already the newest version (1:2.1.6-1build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ traceroute google.com
traceroute to google.com (172.217.26.78), 30 hops max, 60 byte packets
 1 * * *
 2 10.56.0.1 (10.56.0.1) 9.831 ms 9.202 ms 8.684 ms
 3 161.49.13.38.convergeict.com (161.49.13.38) 15.381 ms * 12.277 ms
 4 161.49.4.240.convergeict.com (161.49.4.240) 11.950 ms 161.49.4.242.convergeict.com (161.49.4.242) 11.383 ms 161.49.4.240.convergeict.com (161.49.4.240) 11.081 ms
 5 161.49.13.151.convergeict.com (161.49.13.151) 18.344 ms * 17.463 ms
 6 161.49.13.181.convergeict.com (161.49.13.181) 9.650 ms 12.159 ms *
 7 * * 161.49.13.183.convergeict.com (161.49.13.183) 15.864 ms
 8 * * lcnld-ak-in-f14.1e100.net (172.217.26.78) 12.725 ms
catlyn@ubuntu-server:~$ _
```

Figure 23. Traceroute results from Server B to google.com, showing the path taken across intermediate hops.

DNS Resolution Test – nslookup

The nslookup command was run on Server B to confirm DNS resolution. The query for google.com returned valid IPv4 and IPv6 addresses, confirming that DNS resolution was functioning correctly on the DHCP-configured server as well.

```
catlyn@ubuntu-server:~$ sudo apt install dnsutils -y
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
bind9-dnsutils is already the newest version (1:9.20.24-1ubuntu0.3).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 33
catlyn@ubuntu-server:~$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.27.142
Name:   google.com
Address: 2404:6800:4017:80a::200e
catlyn@ubuntu-server:~$ _
```

Figure 24. nslookup results on Server B confirming successful DNS resolution for google.com.



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Summary of Results

Test	Server A (Static)	Server B (DHCP)
IP Address Assigned	192.168.1.150/24 (static)	192.168.1.12/24 (DHCP)
Ping to other server	Success – 0% packet loss	Success – 0% packet loss
Ping to google.com	Success – 0% packet loss	Success – 0% packet loss
Traceroute to google.com	Completed – 8 hops	Completed – 8 hops
nslookup google.com	Resolved (IPv4 & IPv6)	Resolved (IPv4 & IPv6)

Conclusion

Server A was successfully configured with a static IP address (192.168.1.150/24) using Netplan, while Server B retained its DHCP configuration and automatically obtained an address (192.168.1.12/24) from the network. Connectivity testing confirmed that both servers could reach each other and access the internet without packet loss. Traceroute confirmed the network path to an external host, and nslookup confirmed that DNS resolution was functioning correctly on both servers. This exercise demonstrates a practical understanding of static and dynamic IP configuration in Ubuntu Server, along with the use of core CLI tools (ping, traceroute, nslookup) for network verification and troubleshooting.